Development Environment Setup

README

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This course uses **Zephyr**, **west**, and a **Python virtual environment**.
The setup is designed so that students can get started with minimal effort.

Overview of the Environment

The workspace root is the GitHub repo (zephyr-esp32s3-workspace), and inside it the zephyr-esp32s3-course directory contains the manifest (west.yml) and all course material. The bootstrap script creates a Python virtual environment, installs west, initializes the workspace using west init -l zephyr-esp32s3-course, and downloads Zephyr and its modules into the workspace. This structure means students can simply clone the repo, run the bootstrap script, and have a full Zephyr environment created automatically.

Inside zephyr-esp32s3-course, the apps/ directory is the main home for the examples. Each app lives in its own subfolder, for example apps/welcome/, with the conventional Zephyr structure: CMakeLists.txt, prj.conf, and src/main.c (plus any extra source files). Device tree overlays or app-specific board tweaks sit in apps//boards/ (as we have with apps/welcome/boards/esp32s3_devkitc.overlay).

1. First-Time Setup (Required Once)

Clone the repo and run the bootstrap script.

Linux / macOS

```
git clone https://github.com/johnosbb/zephyr-esp32s3-course.git
cd zephyr-esp32s3-course
./scripts/bootstrap.sh
```

Windows (PowerShell)

```
git clone https://github.com/johnosbb/zephyr-esp32s3-course.git
cd zephyr-esp32s3-course
.\scripts\bootstrap.ps1
```

2. Installing the Zephyr SDK (Only If Requested)

If the bootstrap script cannot find the SDK, it prints instructions.

Choose one of the installation methods below.

Linux x86_64: Install Zephyr SDK with wget (recommended)

The course examples assume the **Zephyr SDK** is installed in your home directory.

```
# Choose the SDK version you want to use (must be compatible with the Zephyr version in west.yml)
SDK_VERSION=0.17.0

# Download the SDK archive for Linux x86_64
wget "https://github.com/zephyrproject-rtos/sdk-ng/releases/download/v${SDK_VERSION}/zephyr-sdk-${SDK_V

# Extract and move into your home directory
tar xf "zephyr-sdk-${SDK_VERSION}_linux-x86_64.tar.xz"
mv "zephyr-sdk-${SDK_VERSION}" "$HOME/zephyr-sdk"

# Run SDK setup script (no sudo needed for a user-local install)
"$HOME/zephyr-sdk/setup.sh"

# Tell Zephyr to use this SDK (for this terminal session)
export ZEPHYR_TOOLCHAIN_VARIANT=zephyr
export ZEPHYR_SDK_INSTALL_DIR="$HOME/zephyr-sdk"
```

Or more permanently place these in your .bashrc file

```
echo 'export ZEPHYR_TOOLCHAIN_VARIANT=zephyr' >> "$HOME/.bashrc"
echo 'export ZEPHYR_SDK_INSTALL_DIR="$HOME/zephyr-sdk"' >> "$HOME/.bashrc"
```

Then reload your shell configuration in the current terminal:

```
source "$HOME/.bashrc"
```

You can verify they are set with:

```
echo "$ZEPHYR_TOOLCHAIN_VARIANT"
echo "$ZEPHYR_SDK_INSTALL_DIR"
```

Key idea: Consistent setup across Windows and Linux.